TrES-3b

R-0016

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_260525` · created 2026-07-08T07:44:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461185.7643 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.088 +/- 0.057
Transit depth (Rp/Rs)^2	0.77 +/- 1.0 [%]
Orbital Inclination (inc)	80.22 +/- 2.87
Airmass coefficient 1 (a1)	1.41 +/- 0.89
Airmass coefficient 2 (a2)	-0.18 +/- 0.62
Scatter in the residuals of the lightcurve fit is	70.25 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.025 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.088 ± 0.057 vs 0.16309 (deviation 1.32σ)
Duration fit vs published	0.025 d vs 0.0567 d (deviation 1.17σ)
Mid-time O–C	0.4 min
Depth SNR	1.5
Residual scatter	70.25 %

Light curve

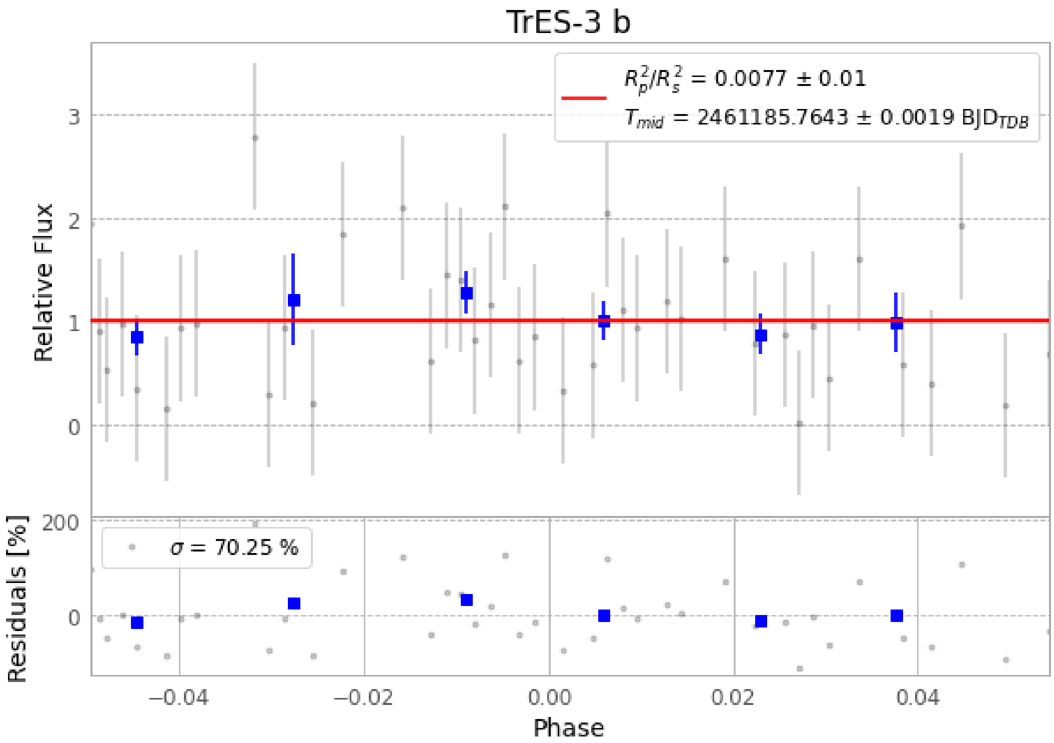

Figure 1 — FinalLightCurve_TrES-3 b_2026-05-24.png (SHA-256 d6186ba1003f3ec6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 309], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4c33aed4e622465f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

67 FITS frames (44 MB), `TRES-3260525044004.FITS` → `TRES-3260525075715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260525.log (7b842f0ac6d2…), FinalLightCurve_TrES-3 b_2026-05-24.png (d6186ba1003f…), FinalLightCurve_TrES-3 b_2026-05-24.csv (74747370a398…)

Compute resources

Wall time 206.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)